

AgroScan Viva Guide

Code map, architecture, and functions with file locations

Study notes for the CSE400 viva

AgroScan — leaf photo → crop → disease → advice

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Chapter 1

What to say first

AgroScan is a Bangladesh farmer decision-support app. A user photographs a leaf. The server asks three questions in order: is this a leaf, which crop is it, which disease is it. Treatment text does not come from the neural net. It comes from a written bilingual pack. Chat uses that pack first. A language model is only for leftover questions. The same API also runs a local shop, a dealer desk, and an admin desk.

One sentence if they interrupt: “It is a gated three-level CNN pipeline plus a written advice pack, served as a Bengali web app.”

Numbers they will ask. 27 crops. About 123 disease classes. Three disease backbones: EfficientNet-B3, ResNet-50, DenseNet-121. Test accuracy about 97.9% / 97.9% / 97.6%. Leaf accept 0.50. Crop confirm 0.90. Uncertain 0.45. Low confidence 0.80. Production: <https://agroscan.mujahidulhaquejihad.com>.

This repo has 434 Python functions and 425 JavaScript functions in `agroscan/`, `backend/`, `scripts/`, and `web/*.js`. You do not memorise all of them. You memorise the ones in Chapter 2 and you can find the rest in the tables.

Chapter 2

Must-know functions

If they ask “where is that in the code?”, these are the answers.

What they ask	Where	What it does
Whole diagnosis	<code>infer.py</code> <code>InferenceEngine.predict</code> 392–472	Leaf → crop → disease, early exits
Is it a leaf?	<code>infer.py</code> <code>_leaf_stage</code> 187– 211	MobileNetV3-Large, threshold 0.50
Which crop?	<code>infer.py</code> <code>_crop_stage</code> 225– 278	EfficientNet-B3, or disease-prob fallback
Which disease?	<code>infer.py</code> <code>_disease_stage</code> 292–389	Mask to crop, max-confidence winner
Mask to crop	<code>infer.py</code> <code>_mask_probs</code> 280– 290	Zero out other crops' classes
HTTP upload	<code>backend/main.py</code> <code>predict</code> 152–180	POST <code>/api/predict</code>
Build a CNN	<code>models.py</code> <code>build_model</code> 11– 42	timm/torchvision backbone + head
Load .pt	<code>models.py</code> <code>load_checkpoint</code> 70–80	arch, class names, weights
Train loop	<code>engine.py</code> <code>train_model</code> 53–	Focal loss, AMP, resume
Advice text	<code>knowledge.py</code> <code>advice_for</code> 365–	Pack lookup by class name
Chat ladder	<code>llm_chat.py</code> <code>chat_reply</code> 932– 1026	clarify → pack → KB → Gemini
RAG fetch	<code>rag.py</code> <code>retrieve</code> 235–	TF-IDF + cosine, $k = 6$
Shop order	<code>shop_db.py</code> <code>create_order</code> 529–	Insert order, pick supplier
Vendor login	<code>shop_db.py</code> <code>vendor_login</code> 1261–	Token for <code>/vendor</code>
Farmer login	<code>auth_store.py</code> <code>login_local</code> 168–	Session token
Click Analyze	<code>web/core.js</code> <code>runAnalyze</code> 750–	Compress image, POST predict
Draw result	<code>web/core.js</code> <code>renderResult</code> 359–	Gate / crop picker / disease card
Chat UI	<code>web/app-extra.js</code> <code>sendChat</code> 945–	POST <code>/api/chat</code>
Shop cart	<code>web/shop.js</code> <code>checkout</code> 809–	POST <code>/api/shop/orders</code>

Chapter 3

How a request travels

1. Browser: `web/index.html` loads `config.js`, `i18n.js`, `core.js`, `app.js`, `app-extra.js`, `shop.js`.
2. User taps capture. `core.js` `runAnalyze` compresses the photo and POSTs to `/api/predict`.
3. `backend/main.py` `predict` reads bytes, `load_rgb_image`, calls `engine.predict`.
4. `InferenceEngine` was created at startup in `_load` (line 102).
5. JSON comes back: `stage1_leaf_gate`, `stage2_leaf_type`, `stage3_disease`.
6. If crop confidence < 0.90 , UI asks the farmer to pick. Then `runAnalyzeWithCrop` (line 831) sends `crop=` again.
7. Advice is already inside `best_answer.advice` from `advice_for`. Chat is a separate POST `/api/chat`.

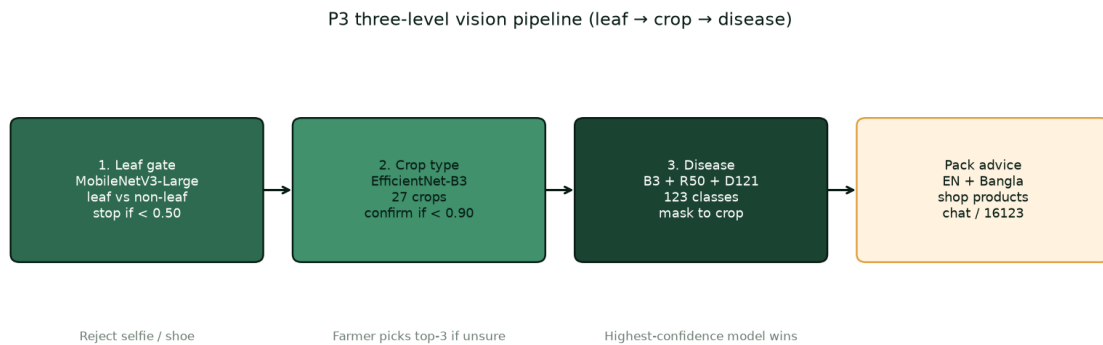


Figure 3.1: Leaf gate, crop, disease, then pack advice.

AgroScan P3 application architecture

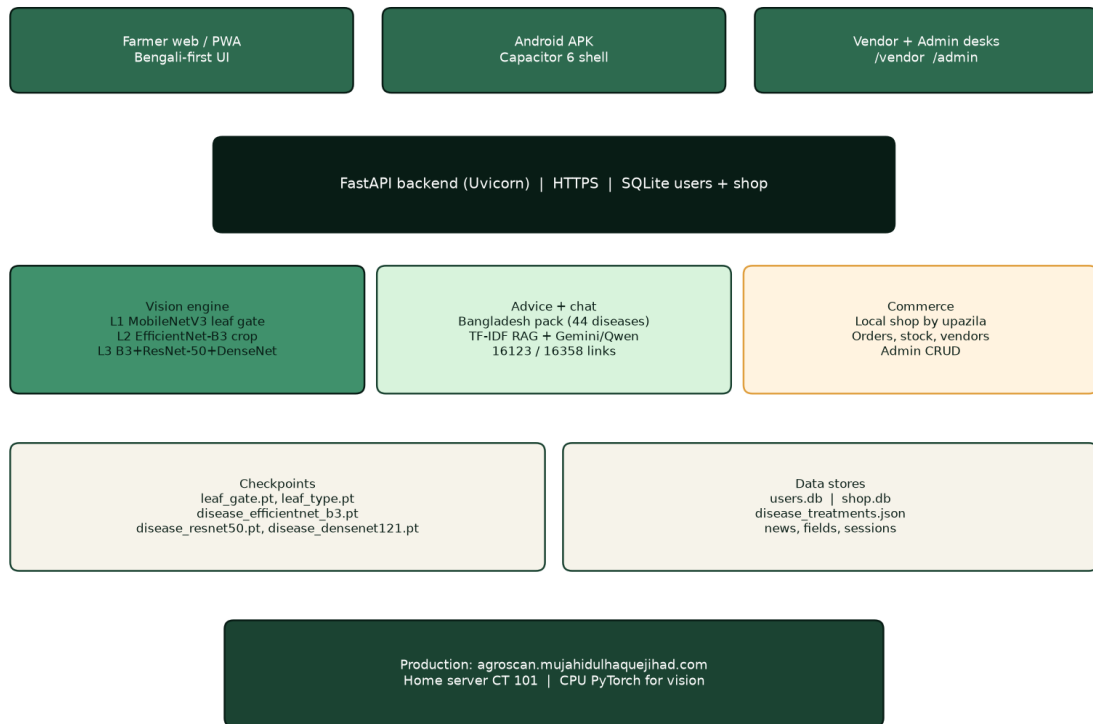


Figure 3.2: Web / APK / vendor / admin all hit the same FastAPI process.

Chapter 4

Vision pipeline (say this slowly)

4.1 Level 1 — leaf gate

File: `agrosan/infer.py`. Model file: `models/leaf_gate.pt`. Architecture: MobileNetV3-Large. Size 224×224 . Classes: `leaf`, `non_leaf`. Threshold `LEAF_ACCEPT_THRESHOLD = 0.5` in `agrosan/config.py` line 162.

`_leaf_stage` (187–211) runs softmax, finds the leaf index, returns `is_leaf`. If false, `predict` returns immediately with a Bangla message (409–413). **Why:** a disease net always picks *some* class. A selfie would become “tomato blight” without the gate.

4.2 Level 2 — crop

File: same. Checkpoint: `models/leaf_type.pt`. EfficientNet-B3. Size 300×300 . 27 crops. `_crop_stage` (225–278). If confidence $< \text{CROP_USER_CONFIRM_THRESHOLD}$ (0.90, config line 164), `needs_user_pick` is set and disease is skipped (445–455). If the crop looks like “Other” (`CROP\_OTHER\_THRESHOLD` 0.55), disease is skipped.

Why crop first: tomato early blight and potato early blight are different pack cards. Masking happens in `_mask_probs`.

4.3 Level 3 — disease ensemble

Three checkpoints from `config.DISEASE_ARCHS`: EfficientNet-B3, ResNet-50, DenseNet-121. About 123 classes named `Crop__Disease`. Each model softmax is aligned onto the largest vocabulary (`_align_probs`, 157–165). Then each vector is masked to the chosen crop. **We do not average.** Line 341: `winner = max(per_model, key=confidence)`. That is the sentence they want.

Flags on the winner: `uncertain` if `conf < 0.45`, `low_confidence` if `conf < 0.80` (`config.py` 158–160). Advice is attached via `advice_for` (380–382).

`_pretty` (70–91) splits `Tomato__Late_blight` into plant + condition.

4.4 Preprocessing

`agrosan/image_io.py load_rgb_image` turns bytes/PIL into RGB. Optional auto-crop: `agrosan/leaf_crop.py auto_crop_leaf` (HSV green mask + largest contour), exposed as

POST /api/preprocess/auto-crop. Transforms: agroscan/data.py infer_transform.

Chapter 5

Training and data

Command: `python -m agrosan.train_all → train_leaf.py, train_crop.py, train_disease.py`.

Shared loop: `engine.train_model`. Loss: `FocalLoss` (engine.py 18–33) if `USE_FOCAL_LOSS`.

Checkpoints: `models/leaf_gate.pt, leaf_type.pt, disease_efficientnet_b3.pt, disease_resnet50.pt, disease_densenet121.pt`. Resume files: `*.train.pt`.

Folders: `Datasets/leaf_gate/, Datasets/leaf_type/, Datasets/train|valid|test/`. Sources: PlantVillage, PlantDoc, Bangladesh sets. `data.py` `disease_loaders / combined_disease_loaders` build the `DataLoaders`.

If they ask why accuracy is high: large test set ($\sim 30k$ images), transfer learning, three similar CNNs. **Honest limit:** test photos are still cleaner than a dark field phone photo. That is why the gate and the 90% crop confirm exist.

Chapter 6

Advice, chat, RAG

6.1 Pack

`data/agroscan/` JSON cards (about 44 diseases, EN+BN). `knowledge.py` `advice_for` maps a vision class name onto a card (pathogen, symptoms, cultural control, chemicals, 16123). Chat formatting: `_format_pack_reply` (599). Name matching: `resolve_disease_query` (803).

6.2 Chat order (memorise)

`llm_chat.chat_reply` (932–1026):

1. Clarify if the name fits several crops (`source: clarify`).
2. Pack catalog / matched pack (`source: pack`) — LLM is not called.
3. Small KB: hello, 16123, how to use the app (`knowledge.chatbot_reply`, 888).
4. If Gemini is on: `generate_reply` with RAG snippets (`source: gemini+rag`).
5. Else optional local Qwen. If the text looks bad, fall back to KB.

6.3 RAG

Not FAISS. Sklearn TF-IDF + cosine in `agroscan/rag.py` `retrieve`. Top 6 chunks. If a disease key is already known, `allow_keys` pins retrieval so rice blast pages cannot leak maize blight doses.

HTTP: POST `/api/chat` → `backend/main.py` `chat` (265). Vision+chat wrap: POST `/api/chat/vision` → `vision_scan_reply` (1166).

Chapter 7

Shop, vendor, admin, accounts

SQLite `shop.db`. Schema and queries: `agrosan/shop_db.py`. Seed from pack: `shop_seed.py seed_from_pack`, called at API startup if the catalogue is thin.

Farmer: GET `/api/shop/products`, POST `/api/shop/orders`. Vendor token $\neq$ farmer token $\neq$ admin token. Vendor may only see their `supplier_id` orders (`vendor_order_detail`, `main.py` 1028). Vendor shop profile: PATCH `/api/vendor/me` (982). Farmer auth: `auth_store.py` + `/api/auth/*`.

Chapter 8

Frontend files (what each JS file is for)

- `config.js` — `API_BASE`, online pill against `/api/status`.
- `i18n.js` — Bangla/English strings.
- `core.js` — status, upload, analyze, render diagnosis.
- `app.js` — page init, older upload/chat wiring.
- `app-extra.js` — chat panel, mic, speech, vision-in-chat.
- `shop.js` — catalogue, cart, checkout, orders.
- `field.js` — named plots.
- `auth.js` / `account.js` — login header and account page.
- `crop.js` — perspective / auto-crop UI.
- `offline-infer.js` — leftover ONNX path; production APK does not ship models.
- `ops.js` — shared helpers for vendor/admin HTML.
- Vendor UI lives in `web/vendor.html` (inline script). Admin in `web/admin.html`.
- Android: `mobile/` Capacitor shell. No on-device ensemble.

Chapter 9

API list you should be able to name

Diagnosis: GET /api/status, POST /api/predict, POST /api/preprocess/auto-crop. Chat: POST /api/chat, POST /api/chat/vision, POST /api/chat/stt. Shop: /api/shop/products, /api/shop/orders. Auth: /api/auth/signup|login|me. Vendor: /api/vendor/login, /me, /orders, /listings. Admin: /api/admin/login, users, orders, products, vendors. HTML pages: /, /vendor, /admin. Every route function is in Table 11.4.

Chapter 10

Likely viva questions

1. **Why three levels?** Disease nets always output a class. Gate stops non-leaves. Crop locking stops tomato spots being called rice blast.
2. **Why not average the three disease models?** A weak model would dilute a confident one. Code takes the max confidence (`infer.py` 341).
3. **Why is advice not from the CNN?** CNNs see pixels. Doses and 16123 come from a curated pack (`advice_for`).
4. **What if crop confidence is 70%?** Farmer must pick from top-3. Disease does not run yet.
5. **Bengali?** Default UI language. Pack and chat have BN strings. `i18n.js`.
6. **Can it work offline?** Shell/PWA cache the UI. Diagnosis needs the server. ONNX was removed from the APK.
7. **Animal disease?** In the literature, not in the running app. Say that clearly.
8. **Grad-CAM?** Used in the model-comparison notebook, not on every farmer card. Trust in production is gate + crop confirm + pack + helpline.
9. **Imbalance?** Focal loss + balanced sampler in `engine.py` / `data.py`.
10. **Security?** Separate tokens; vendor scoped to supplier; hashed passwords; no mixing farmer and admin sessions.

Chapter 11

Python function index

Open the file at the line in the right-hand column. Nested helpers inside a function are not listed; only top-level functions and class methods.

Table 11.1: Vision engine, training loop, and image helpers.

File	Function	Lines
agroscan/config.py	_split_has_classes	54–55
agroscan/config.py	disease_ckpt	106–107
agroscan/config.py	model_display_name	178–179
agroscan/config.py	load_dotenv	182–199
agroscan/data.py	_folder_image_loader	26–28
agroscan/data.py	build_transforms	34–54
agroscan/data.py	infer_transform	57–58
agroscan/data.py	_dataloader_kwargs	61–72
agroscan/data.py	ImageListDataset.__init__	79–81
agroscan/data.py	ImageListDataset.__len__	83–84
agroscan/data.py	ImageListDataset.__getitem__	86–89
agroscan/data.py	_gather	92–93
agroscan/data.py	_class_image_paths	96–106
agroscan/data.py	_stratified_split	109–126
agroscan/data.py	_items_from_imagefolder_root	129–137
agroscan/data.py	_plantvillage_class_names	140–145
agroscan/data.py	_plantvillage_only_loaders	151–167
agroscan/data.py	_combined_disease_items	170–252
agroscan/data.py	DiseaseImageFolder.__init__	258–265
agroscan/data.py	DiseaseImageFolder.find_classes	268–280
agroscan/data.py	class_weights_for_dataset	283–290
agroscan/data.py	_make_loader	293–305
agroscan/data.py	_unified_split_loaders	308–320
agroscan/data.py	crop_loaders	323–337
agroscan/data.py	combined_disease_loaders	340–367
agroscan/data.py	disease_loaders	370–399
agroscan/data.py	disease_image_paths_for_leaf_gate	402–443
agroscan/data.py	_leaf_items	449–473
agroscan/data.py	leaf_loaders	476–516
agroscan/engine.py	FocalLoss.__init__	21–27
agroscan/engine.py	FocalLoss.forward	29–33
agroscan/engine.py	evaluate	37–45
agroscan/engine.py	train_resume_path	48–50
agroscan/engine.py	train_model	53–188
agroscan/image_io.py	openable_path	12–19
agroscan/image_io.py	_downscale_if_needed	72–79
agroscan/image_io.py	_to_rgb	82–96
agroscan/image_io.py	load_rgb_image	99–139

File	Function	Lines
agroscan/image_io.py	sniff_image_path	142–157
agroscan/image_io.py	is_image_path	160–168
agroscan/image_io.py	gather_images	171–179
agroscan/infer.py	_ensure_other_option	27–32
agroscan/infer.py	_decorate_crop	35–53
agroscan/infer.py	_other_crop_message	56–67
agroscan/infer.py	_pretty	70–91
agroscan/infer.py	LoadedModel.__init__	95–100
agroscan/infer.py	LoadedModel.probs	103–105
agroscan/infer.py	InferenceEngine.__init__	109–134
agroscan/infer.py	InferenceEngine.ready	138–139
agroscan/infer.py	InferenceEngine.status	141–154
agroscan/infer.py	InferenceEngine._align_probs	157–165
agroscan/infer.py	InferenceEngine._disease_prob_stack	167–185
agroscan/infer.py	InferenceEngine._leaf_stage	187–211
agroscan/infer.py	InferenceEngine._top_crops_from_probs	213–223
agroscan/infer.py	InferenceEngine._crop_stage	225–278
agroscan/infer.py	InferenceEngine._mask_probs	280–290
agroscan/infer.py	InferenceEngine._disease_stage	292–389
agroscan/infer.py	InferenceEngine.predict	392–472
agroscan/label_map.py	merge_class_name	73–76
agroscan/label_map.py	is_variety_class	79–80
agroscan/label_map.py	is_other_crop	83–85
agroscan/label_map.py	crop_from_class	88–103
agroscan/label_map.py	classes_for_crop	106–114
agroscan/label_map.py	merge_key	117–135
agroscan/label_map.py	_bd_to_canonical	138–148
agroscan/label_map.py	_mango_variety_to_canonical	151–155
agroscan/label_map.py	_bd_leaf_to_canonical	158–163
agroscan/label_map.py	canonical_name	166–201
agroscan/label_map.py	LabelRegistry.__init__	207–211
agroscan/label_map.py	LabelRegistry.class_names	214–215
agroscan/label_map.py	LabelRegistry.register	217–225
agroscan/label_map.py	LabelRegistry.label_for	227–228
agroscan/label_map.py	summarize_registry	231–238
agroscan/land_scale.py	to_decimals	8–23
agroscan/land_scale.py	_first_range	26–41
agroscan/land_scale.py	water_litres_per_decimal	44–55
agroscan/land_scale.py	dose_per_litre	58–112
agroscan/land_scale.py	kg_per_bigha	115–143
agroscan/land_scale.py	land_band	146–153
agroscan/land_scale.py	followup_steps	156–241
agroscan/land_scale.py	scale_chemical	244–317
agroscan/land_scale.py	scale_fertilizer_note	320–338
agroscan/land_scale.py	land_summary	341–354
agroscan/leaf_crop.py	_pil_to_bgr	20–22
agroscan/leaf_crop.py	_bgr_to_pil	25–27
agroscan/leaf_crop.py	_leaf_mask	30–60
agroscan/leaf_crop.py	_largest_contour	63–67
agroscan/leaf_crop.py	auto_crop_leaf	76–144
agroscan/leaf_crop.py	perspective_crop	147–175
agroscan/leaf_crop.py	image_to_jpeg_bytes	178–181
agroscan/models.py	build_model	11–41
agroscan/models.py	save_checkpoint	44–53
agroscan/models.py	save_training_state	56–58
agroscan/models.py	load_training_state	61–67
agroscan/models.py	load_checkpoint	70–80

Table 11.2: Chat, pack, RAG, and advice.

File	Function	Lines
agrosan/farm_plan.py	season_for_month	22–27
agrosan/farm_plan.py	load_flora_crops	75–94
agrosan/farm_plan.py	load_csv_crop_rows	98–116
agrosan/farm_plan.py	_land_note	119–138
agrosan/farm_plan.py	recommend_crops	141–299
agrosan/farm_plan.py	cultivation_outline	302–370
agrosan/knowledge.py	_attach_guide	297–309
agrosan/knowledge.py	_localize_info	312–326
agrosan/knowledge.py	advice_for_key	329–362
agrosan/knowledge.py	advice_for	365–386
agrosan/knowledge.py	all_diseases	389–409
agrosan/knowledge.py	_contacts_text	422–430
agrosan/knowledge.py	_is_healthy_rec	469–472
agrosan/knowledge.py	wants_catalog	475–485
agrosan/knowledge.py	_catalog_reply_lang	488–496
agrosan/knowledge.py	pack_catalog_reply	499–575
agrosan/knowledge.py	_split_name	578–579
agrosan/knowledge.py	_disease_name_tokens	582–596
agrosan/knowledge.py	_format_pack_reply	599–634
agrosan/knowledge.py	_option_label	637–642
agrosan/knowledge.py	_score_pack_candidates	645–725
agrosan/knowledge.py	_score_kb_candidates	728–771
agrosan/knowledge.py	_clarify_payload	774–800
agrosan/knowledge.py	resolve_disease_query	803–867
agrosan/knowledge.py	resolve_disease_advice	870–879
agrosan/knowledge.py	_suggestions	882–885
agrosan/knowledge.py	chatbot_reply	888–1014
agrosan/llm_chat.py	_env_flag	63–64
agrosan/llm_chat.py	_adapter_path	67–74
agrosan/llm_chat.py	_gemini_key	77–81
agrosan/llm_chat.py	_gemini_enabled	84–88
agrosan/llm_chat.py	_want_local_llm	91–92
agrosan/llm_chat.py	_local_base_id	95–96
agrosan/llm_chat.py	_local_provider	99–105
agrosan/llm_chat.py	llm_configured	108–123
agrosan/llm_chat.py	llm_status	126–146
agrosan/llm_chat.py	_structural_intent	149–170
agrosan/llm_chat.py	_kb_is_generic	173–177
agrosan/llm_chat.py	_relevant_context	180–215
agrosan/llm_chat.py	_sanitize_history	231–255
agrosan/llm_chat.py	_pack_keys	258–265
agrosan/llm_chat.py	_llm_looks_unsure	268–314
agrosan/llm_chat.py	_gemini_contents	317–336
agrosan/llm_chat.py	_reply_lang	339–353
agrosan/llm_chat.py	_build_chat_messages	356–469
agrosan/llm_chat.py	_fold_system_into_user	472–484
agrosan/llm_chat.py	_to_chat_prompt	487–498
agrosan/llm_chat.py	_resolve_device	501–509
agrosan/llm_chat.py	_want_4bit	512–518
agrosan/llm_chat.py	_load_base	521–558
agrosan/llm_chat.py	_ensure_loaded	561–638
agrosan/llm_chat.py	preload	641–651
agrosan/llm_chat.py	_generate_text	654–681
agrosan/llm_chat.py	_gemini_models	684–693
agrosan/llm_chat.py	_gemini_extract_text	696–698
agrosan/llm_chat.py	_gemini_extract	701–709
agrosan/llm_chat.py	_gemini_http	712–725
agrosan/llm_chat.py	_gemini_generate	728–813
agrosan/llm_chat.py	generate_reply	816–836
agrosan/llm_chat.py	transcribe_audio	842–929

File	Function	Lines
agrosan/llm_chat.py	chat_reply	932–1026
agrosan/llm_chat.py	_scan_header	1029–1046
agrosan/llm_chat.py	_build_vision_messages	1049–1108
agrosan/llm_chat.py	_llm_text_is_bad	1111–1137
agrosan/llm_chat.py	generate_vision_feedback	1140–1163
agrosan/llm_chat.py	vision_scan_reply	1166–1310
agrosan/news.py	_load_raw	14–21
agrosan/news.py	_pick	24–28
agrosan/news.py	_seasonal_extra	31–70
agrosan/news.py	list_news	73–127
agrosan/rag.py	_section_bodies	29–64
agrosan/rag.py	_guide_text	67–78
agrosan/rag.py	_info_text	81–90
agrosan/rag.py	_load_pack_docs	93–109
agrosan/rag.py	_pack_chunks	112–141
agrosan/rag.py	_legacy_chunks	144–167
agrosan/rag.py	_chunks	170–176
agrosan/rag.py	clear_index_cache	179–180
agrosan/rag.py	_index	184–193
agrosan/rag.py	_key_ok	196–218
agrosan/rag.py	_section_boost	221–232
agrosan/rag.py	retrieve	235–289
agrosan/rag_build.py	_is_healthy	19–22
agrosan/rag_build.py	_lines	25–48
agrosan/rag_build.py	_chem_block	51–98
agrosan/rag_build.py	_pick	101–104
agrosan/rag_build.py	build_disease_doc	107–230
agrosan/rag_build.py	build_all	233–251
agrosan/treatment_detail.py	_pick	21–24
agrosan/treatment_detail.py	_warnings_for	27–81
agrosan/treatment_detail.py	_chem_display	84–111
agrosan/treatment_detail.py	treatment_plan_for	114–124
agrosan/treatment_detail.py	_plan_from_record	127–164
agrosan/treatment_detail.py	advice_from_pack	167–238

Table 11.3: Shop database and farmer authentication.

File	Function	Lines
agrosan/auth_store.py	_connect	16–20
agrosan/auth_store.py	_ensure_delivery_columns	23–36
agrosan/auth_store.py	init_db	39–63
agrosan/auth_store.py	_hash_password	66–69
agrosan/auth_store.py	_check_password	72–78
agrosan/auth_store.py	_user_dict	81–101
agrosan/auth_store.py	_session_counts	104–117
agrosan/auth_store.py	_new_token	120–121
agrosan/auth_store.py	_expires_at	124–125
agrosan/auth_store.py	create_local_user	128–165
agrosan/auth_store.py	login_local	168–178
agrosan/auth_store.py	get_user_by_token	181–198
agrosan/auth_store.py	upsert_google_user	201–220
agrosan/auth_store.py	create_session	223–231
agrosan/auth_store.py	delete_session	234–239
agrosan/auth_store.py	update_user_profile	242–278
agrosan/auth_store.py	admin_list_users	281–303
agrosan/auth_store.py	admin_get_user	306–313
agrosan/auth_store.py	admin_update_user	316–399
agrosan/auth_store.py	admin_revoke_user_sessions	402–407

File	Function	Lines
agrosan/auth_store.py	admin_delete_user	410–418
agrosan/auth_store.py	admin_user_stats	421–444
agrosan/shop_db.py	_connect	28–33
agrosan/shop_db.py	init_shop_db	36–164
agrosan/shop_db.py	_ensure_columns	167–205
agrosan/shop_db.py	product_price	208–223
agrosan/shop_db.py	update_product_price	226–243
agrosan/shop_db.py	list_products_admin	246–247
agrosan/shop_db.py	_now	250–251
agrosan/shop_db.py	_row_dict	254–255
agrosan/shop_db.py	list_upazilas	258–290
agrosan/shop_db.py	_enrich_upazila_bn	293–319
agrosan/shop_db.py	_haversine_km	321–328
agrosan/shop_db.py	nearest_upazila	331–351
agrosan/shop_db.py	list_suppliers	354–392
agrosan/shop_db.py	list_products	395–436
agrosan/shop_db.py	product_with_suppliers	439–460
agrosan/shop_db.py	recommended_products_for_disease	463–522
agrosan/shop_db.py	_order_code	525–526
agrosan/shop_db.py	create_order	529–617
agrosan/shop_db.py	get_order	620–644
agrosan/shop_db.py	list_orders	647–685
agrosan/shop_db.py	admin_shop_stats	688–717
agrosan/shop_db.py	update_order_status	720–747
agrosan/shop_db.py	update_order	750–839
agrosan/shop_db.py	pick_supplier_id	842–853
agrosan/shop_db.py	assign_order_supplier	856–873
agrosan/shop_db.py	list_suppliers_admin	876–900
agrosan/shop_db.py	get_supplier	903–908
agrosan/shop_db.py	list_supplier_listings	911–943
agrosan/shop_db.py	update_supplier_listing	946–986
agrosan/shop_db.py	upsert_supplier_listing	989–1022
agrosan/shop_db.py	update_supplier_profile	1025–1041
agrosan/shop_db.py	vendor_self_update	1044–1068
agrosan/shop_db.py	_vendor_username	1071–1075
agrosan/shop_db.py	_vendor_dict	1078–1092
agrosan/shop_db.py	_ensure_default_vendor	1095–1141
agrosan/shop_db.py	list_vendors	1144–1162
agrosan/shop_db.py	create_vendor	1165–1184
agrosan/shop_db.py	reset_vendor_password	1187–1201
agrosan/shop_db.py	update_vendor	1204–1245
agrosan/shop_db.py	set_vendor_active	1248–1258
agrosan/shop_db.py	vendor_login	1261–1282
agrosan/shop_db.py	vendor_logout	1285–1289
agrosan/shop_db.py	get_vendor_by_token	1292–1315
agrosan/shop_db.py	vendor_stats	1318–1354
agrosan/shop_seed.py	_product_image_map	14–22
agrosan/shop_seed.py	_short_desc_map	25–33
agrosan/shop_seed.py	_image_url_for	36–41
agrosan/shop_seed.py	_short_description	44–63
agrosan/shop_seed.py	seed_from_pack	66–255
agrosan/shop_seed.py	apply_product_images	258–298

Table 11.4: FastAPI routes and helpers in backend/main.py.

File	Function	Lines
backend/main.py	_optional_user	95–98
backend/main.py	_load	102–123

File	Function	Lines
backend/main.py	status	127–133
backend/main.py	llm_load	137–144
backend/main.py	classes	148–149
backend/main.py	predict	153–182
backend/main.py	preprocess_auto_crop	190–206
backend/main.py	preprocess_perspective_crop	210–225
backend/main.py	chat_stt	249–262
backend/main.py	chat	266–286
backend/main.py	chat_vision	290–322
backend/main.py	resources	326–327
backend/main.py	news	331–333
backend/main.py	advice	337–352
backend/main.py	diseases	356–359
backend/main.py	crops_catalog	375–378
backend/main.py	farm_plan	382–395
backend/main.py	farm_cultivate	399–402
backend/main.py	shop_upazilas	483–484
backend/main.py	shop_nearest_upazila	488–492
backend/main.py	shop_suppliers	496–507
backend/main.py	shop_products	511–521
backend/main.py	shop_product	525–529
backend/main.py	shop_recommend	533–534
backend/main.py	shop_create_order	538–555
backend/main.py	shop_orders	559–562
backend/main.py	shop_order_detail	566–573
backend/main.py	_admin_expected_token	579–580
backend/main.py	_admin_expected_user	583–584
backend/main.py	_admin_token_ok	587–593
backend/main.py	_vendor_user	596–602
backend/main.py	admin_login	606–611
backend/main.py	admin_orders	615–627
backend/main.py	admin_order_detail	631–637
backend/main.py	admin_stats	641–656
backend/main.py	admin_users	660–668
backend/main.py	admin_user_detail	672–678
backend/main.py	admin_user_update	693–714
backend/main.py	admin_user_create	718–738
backend/main.py	admin_user_revoke	742–746
backend/main.py	admin_user_delete	750–757
backend/main.py	admin_update_order	761–797
backend/main.py	admin_create_order	801–823
backend/main.py	admin_seed	827–832
backend/main.py	admin_apply_images	836–841
backend/main.py	admin_products	844–857
backend/main.py	_apply_admin_product_price	860–864
backend/main.py	admin_set_product_price	868–877
backend/main.py	admin_update_product_price	881–908
backend/main.py	admin_suppliers	912–920
backend/main.py	admin_vendors_list	924–927
backend/main.py	admin_vendors_create	931–938
backend/main.py	admin_vendors_patch	942–959
backend/main.py	vendor_login_api	963–967
backend/main.py	vendor_logout_api	971–974
backend/main.py	vendor_me	978–979
backend/main.py	vendor_me_patch	983–1003
backend/main.py	vendor_stats_api	1007–1008
backend/main.py	vendor_orders_api	1012–1024
backend/main.py	vendor_order_detail	1028–1035
backend/main.py	vendor_update_order	1039–1052
backend/main.py	vendor_listings	1056–1069
backend/main.py	vendor_listing_create	1073–1083
backend/main.py	vendor_listing_patch	1087–1097
backend/main.py	safety_ref	1101–1111

File	Function	Lines
backend/main.py	_auth_response	1137–1139
backend/main.py	auth_signup	1143–1158
backend/main.py	auth_login	1162–1167
backend/main.py	auth_logout	1171–1174
backend/main.py	auth_me	1178–1181
backend/main.py	auth_profile	1193–1207
backend/main.py	auth_google	1211–1232

Table 11.5: Training entrypoints, dataset prep, and helper scripts.

File	Function	Lines
agroscan/bd_data.py	_read	12–16
agroscan/bd_data.py	treatments	20–22
agroscan/bd_data.py	treatments_by_class	26–27
agroscan/bd_data.py	treatments_by_kb	31–38
agroscan/bd_data.py	products	42–44
agroscan/bd_data.py	equipment	48–50
agroscan/bd_data.py	catalogue_by_sku	54–60
agroscan/bd_data.py	suppliers	64–66
agroscan/bd_data.py	upazilas	70–72
agroscan/bd_data.py	crops	76–78
agroscan/bd_data.py	safety	82–84
agroscan/bd_data.py	warning_boxes	88–90
agroscan/bd_data.py	warning_box	93–109
agroscan/bd_data.py	_norm	112–121
agroscan/bd_data.py	find_treatment	124–158
agroscan/bd_data.py	find_crop	161–172
agroscan/bd_dataset.py	_require_datasets	24–31
agroscan/bd_dataset.py	hf_repo_id	34–35
agroscan/bd_dataset.py	hf_access_help	38–45
agroscan/bd_dataset.py	load_bd_splits	48–68
agroscan/bd_dataset.py	_label_names_from_train	71–77
agroscan/bd_dataset.py	_row_label_name	80–83
agroscan/bd_dataset.py	bd_label_names	86–89
agroscan/bd_dataset.py	bd_split_stats	92–94
agroscan/bd_dataset.py	export_to_imagefolder	97–127
agroscan/bd_dataset.py	_safe_class_dir	130–132
agroscan/bd_dataset.py	print_dataset_info	135–148
agroscan/bd_leaf_dataset.py	find_bd_leaf_zip	15–20
agroscan/bd_leaf_dataset.py	is_bd_leaf_extracted	23–30
agroscan/bd_leaf_dataset.py	ensure_bd_leaf_extracted	33–78
agroscan/bd_leaf_dataset.py	bd_leaf_stats	81–89
agroscan/disease_guides.py	guide_for	514–515
agroscan/prepare_bd_dataset.py	_write_manifest	30–42
agroscan/prepare_bd_dataset.py	main	45–95
agroscan/prepare_bd_leaf_dataset.py	main	16–44
agroscan/prepare_hierarchy.py	_unique_path	18–28
agroscan/prepare_hierarchy.py	_safe_name	31–32
agroscan/prepare_hierarchy.py	_hardlink	35–42
agroscan/prepare_hierarchy.py	_iter_images	45–48
agroscan/prepare_hierarchy.py	merge_duplicate_labels	51–80
agroscan/prepare_hierarchy.py	build_leaf_type	83–107
agroscan/prepare_hierarchy.py	drop_variety_from_disease	110–124
agroscan/prepare_hierarchy.py	relocate_leaf_gate	127–140
agroscan/prepare_hierarchy.py	main	143–158
agroscan/train_all.py	main	12–43
agroscan/train_crop.py	train_crop	15–44
agroscan/train_crop.py	main	47–59

File	Function	Lines
agroscan/train_disease.py	train_one	15–41
agroscan/train_disease.py	main	44–57
agroscan/train_leaf.py	train_leaf	15–43
agroscan/train_leaf.py	main	46–58
scripts/build_shop_images.py	load_font	48–60
scripts/build_shop_images.py	wrap	63–82
scripts/build_shop_images.py	short_desc	85–111
scripts/build_shop_images.py	palette_for	114–127
scripts/build_shop_images.py	draw_pack	130–167
scripts/build_shop_images.py	render_card	170–234
scripts/build_shop_images.py	load_rows	237–247
scripts/build_shop_images.py	main	250–274
scripts/check_auto_crop_fast.py	main	12–23
scripts/export_mobile_onnx.py	_slim_advice	34–52
scripts/export_mobile_onnx.py	_export_one	55–89
scripts/export_mobile_onnx.py	main	92–132
scripts/fetch_shop_images.py	_get	89–92
scripts/fetch_shop_images.py	clean_name	95–106
scripts/fetch_shop_images.py	primary_key	109–119
scripts/fetch_shop_images.py	search_terms	122–150
scripts/fetch_shop_images.py	openverse_search	153–172
scripts/fetch_shop_images.py	pick_from_results	175–189
scripts/fetch_shop_images.py	pick_image	195–215
scripts/fetch_shop_images.py	download_image	218–237
scripts/fetch_shop_images.py	load_catalog	240–250
scripts/fetch_shop_images.py	file_key	253–256
scripts/fetch_shop_images.py	main	259–337
scripts/fix_split_leakage.py	log	32–33
scripts/fix_split_leakage.py	orig_name	36–39
scripts/fix_split_leakage.py	pv_source	42–49
scripts/fix_split_leakage.py	file_id_entry	52–57
scripts/fix_split_leakage.py	file_id	60–65
scripts/fix_split_leakage.py	iter_images	68–77
scripts/fix_split_leakage.py	class_counts	80–93
scripts/fix_split_leakage.py	keep_best_split	96–103
scripts/fix_split_leakage.py	scan_leaks	106–149
scripts/fix_split_leakage.py	protect_last_train	152–171
scripts/fix_split_leakage.py	inode_index	174–188
scripts/fix_split_leakage.py	cap_train	191–209
scripts/fix_split_leakage.py	summarize	212–222
scripts/fix_split_leakage.py	main	225–296
scripts/make_app_icons.py	_key_black	49–61
scripts/make_app_icons.py	_trim	64–66
scripts/make_app_icons.py	_paste_centered	69–82
scripts/make_app_icons.py	_square	85–89
scripts/make_app_icons.py	_save	92–95
scripts/make_app_icons.py	main	98–154
scripts/scan_split_leakage.py	log	17–18
scripts/scan_split_leakage.py	orig_name	21–24
scripts/scan_split_leakage.py	pv_source	27–35
scripts/scan_split_leakage.py	scan_task	38–93
scripts/scan_split_leakage.py	main	96–127
scripts/test_rag_chunking.py	main	11–47
scripts/test_split_leakage.py	main	9–27

Chapter 12

JavaScript function index

These are `function name(` declarations inside `web/*.js`. Vendor and admin logic that sits in the HTML script tags is not in this list; read `web/vendor.html` and `web/admin.html` for those desks.

Table 12.1: JavaScript functions in `web/*.js`.

File	Function	Lines
web/account.js	T	7
web/account.js	esc	10
web/account.js	session	15
web/account.js	token	18
web/account.js	isRealUser	22
web/account.js	xhr	27
web/account.js	fmtWhen	46
web/account.js	statusLabel	57
web/account.js	timelineHtml	63
web/account.js	renderOrderCard	82
web/account.js	renderDetail	94
web/account.js	loadOrderDetail	132
web/account.js	setTab	144
web/account.js	dash	157
web/account.js	fillProfile	165
web/account.js	setEditMode	191
web/account.js	cancelEdit	208
web/account.js	saveProfile	220
web/account.js	loadOrders	250
web/account.js	showApp	287
web/account.js	showGate	304
web/account.js	init	309
web/ads.js	t	13
web/ads.js	setStrip	19
web/ads.js	nextStrip	40
web/ads.js	startRotate	45
web/ads.js	stopRotate	51
web/ads.js	closePopup	58
web/ads.js	showPopup	64
web/ads.js	boot	74
web/app-extra.js	init	7
web/app-extra.js	i18n	35
web/app-extra.js	catLabel	37
web/app-extra.js	visitLabel	42
web/app-extra.js	esc	46
web/app-extra.js	renderGovCard	50
web/app-extra.js	renderEmergencyCard	65

File	Function	Lines
web/app-extra.js	paintEmergency	76
web/app-extra.js	filterGovLinks	84
web/app-extra.js	paintGovGrid	97
web/app-extra.js	setupGovHub	107
web/app-extra.js	loadResources	127
web/app-extra.js	setupDragDrop	149
web/app-extra.js	setupChat	164
web/app-extra.js	tChat	229
web/app-extra.js	isBnChat	234
web/app-extra.js	chatSourceLabel	257
web/app-extra.js	rememberChatDisease	267
web/app-extra.js	pushChatBot	273
web/app-extra.js	collectChatHistory	294
web/app-extra.js	pushChatUserImage	314
web/app-extra.js	postChatVision	336
web/app-extra.js	chatAnalyzeImage	370
web/app-extra.js	handleChatVisionResult	396
web/app-extra.js	refreshMicTitle	473
web/app-extra.js	refreshSpeakTitle	482
web/app-extra.js	chatLangTag	492
web/app-extra.js	pickChatVoice	496
web/app-extra.js	stopChatSpeak	517
web/app-extra.js	speakChat	526
web/app-extra.js	toggleChatSpeak	557
web/app-extra.js	stopRecStream	571
web/app-extra.js	stopVoiceListen	587
web/app-extra.js	sendHeardChat	604
web/app-extra.js	recMimeType	612
web/app-extra.js	showSttBusy	624
web/app-extra.js	hideSttBusy	630
web/app-extra.js	encodeWavMono16	634
web/app-extra.js	wstr	638
web/app-extra.js	blobToSpeechWav	662
web/app-extra.js	decode	672
web/app-extra.js	uploadStt	718
web/app-extra.js	beginMediaListen	760
web/app-extra.js	beginVoiceListen	831
web/app-extra.js	toggleVoiceListen	835
web/app-extra.js	stopChatLoading	854
web/app-extra.js	startChatLoading	861
web/app-extra.js	finishChatLoading	881
web/app-extra.js	setChatInputLocked	887
web/app-extra.js	renderChatChips	911
web/app-extra.js	sendChat	945
web/app.js	isImageFile	11
web/app.js	apiDetail	17
web/app.js	initApp	24
web/app.js	setStatusText	48
web/app.js	showOffline	53
web/app.js	showOnline	65
web/app.js	checkStatus	82
web/app.js	loadResources	103
web/app.js	updateAnalyzeButton	121
web/app.js	setupUpload	125
web/app.js	handleFile	141
web/app.js	setupAnalyze	160
web/app.js	render	197
web/app.js	setupChat	302
web/app.js	pushMsg	319
web/app.js	renderChips	328
web/app.js	sendChat	334
web/app.js	setupGoogle	350
web/app.js	decodeJwt	351

File	Function	Lines
web/app.js	showUser	355
web/app.js	onGoogleCredential	361
web/app.js	initGoogle	376
web/auth.js	T	9
web/auth.js	page	13
web/auth.js	getSession	17
web/auth.js	setSession	22
web/auth.js	clearSession	27
web/auth.js	authHeaders	32
web/auth.js	xhr.Json	38
web/auth.js	renderHeader	57
web/auth.js	logout	80
web/auth.js	loginGuest	89
web/auth.js	saveAuthResponse	97
web/auth.js	loginEmail	105
web/auth.js	signupEmail	116
web/auth.js	initPasswordToggles	131
web/auth.js	decodeJwt	151
web/auth.js	googleCredential	156
web/auth.js	initGoogleButton	177
web/auth.js	tryRender	190
web/auth.js	initAuthPage	211
web/auth.js	rewriteStaticAuthLinks	257
web/auth.js	boot	275
web/config.js	isCapacitor	6
web/config.js	isNativeShell	18
web/config.js	apiBase	34
web/config.js	appPath	41
web/config.js	connLabel	60
web/config.js	setConn	67
web/config.js	pingConn	78
web/config.js	bootConn	90
web/core.js	T	10
web/core.js	getLandInputs	16
web/core.js	saveLandInputs	37
web/core.js	syncLandUiFromStorage	49
web/core.js	refreshAdviceForLand	60
web/core.js	setStatus	86
web/core.js	xhrGet	94
web/core.js	formatErr	109
web/core.js	xhrPostFile	117
web/core.js	compressForUpload	159
web/core.js	isImage	210
web/core.js	hasOfflineVision	216
web/core.js	networkLooksUp	220
web/core.js	preferServer	227
web/core.js	updateAnalyzeBtn	232
web/core.js	refreshStatusText	237
web/core.js	checkStatus	241
web/core.js	acceptFile	318
web/core.js	pct	357
web/core.js	renderResult	359
web/core.js	setLoaderText	723
web/core.js	showChecking	736
web/core.js	hideChecking	744
web/core.js	runAnalyze	750
web/core.js	finishOk	786
web/core.js	finishErr	795
web/core.js	runLocal	803
web/core.js	runAnalyzeWithCrop	831
web/core.js	init	848
web/crop.js	checkStageCopy	19
web/crop.js	applyCheckStage	26

File	Function	Lines
web/crop.js	T	34
web/crop.js	ensureOverlay	38
web/crop.js	stopCheckStages	61
web/crop.js	showScan	68
web/crop.js	hideScan	82
web/crop.js	startCheckStages	91
web/crop.js	setBusy	109
web/crop.js	ensureModal	124
web/crop.js	refreshLabels	158
web/crop.js	defaultPoints	170
web/crop.js	draw	180
web/crop.js	hitTest	207
web/crop.js	canvasPos	217
web/crop.js	onDown	228
web/crop.js	onMove	234
web/crop.js	onUp	246
web/crop.js	onTouchStart	247
web/crop.js	onTouchMove	248
web/crop.js	fitCanvas	250
web/crop.js	loadFileToCanvas	266
web/crop.js	blobToFile	275
web/crop.js	loadImg	283
web/crop.js	canvasToFile	290
web/crop.js	shrinkFile	300
web/crop.js	hsv255	316
web/crop.js	isLeafPixel	328
web/crop.js	localAutoCrop	337
web/crop.js	revealModal	394
web/crop.js	open	404
web/crop.js	close	432
web/crop.js	finish	441
web/crop.js	skipOriginal	447
web/crop.js	resetToOriginal	451
web/crop.js	scalePointsToImage	460
web/crop.js	runAutoCrop	472
web/crop.js	applyCrop	494
web/farmer-ux.js	T	8
web/farmer-ux.js	isBn	11
web/farmer-ux.js	currentSeasonTip	25
web/farmer-ux.js	renderSeasonTip	34
web/farmer-ux.js	renderScoutReminder	44
web/farmer-ux.js	cacheAdvice	67
web/farmer-ux.js	loadCachedAdvice	82
web/farmer-ux.js	showOfflineAdvice	89
web/farmer-ux.js	showPhiReminder	173
web/farmer-ux.js	showActionStrip	194
web/farmer-ux.js	wireActionStrip	200
web/farmer-ux.js	wireFeedback	222
web/farmer-ux.js	onDiagnosis	251
web/farmer-ux.js	init	294
web/features.js	bind	8
web/features.js	applyTheme	15
web/features.js	initDark	20
web/features.js	persistHistory	38
web/features.js	saveHistory	68
web/features.js	setHistoryViewMode	75
web/features.js	clearHistoryAll	93
web/features.js	fntHistTime	99
web/features.js	escHtml	102
web/features.js	pctHist	107
web/features.js	listItems	111
web/features.js	compressImageSrc	117
web/features.js	cloneAdvice	154

File	Function	Lines
web/features.js	findUserFeedback	177
web/features.js	buildHistoryEntry	193
web/features.js	renderHistory	270
web/features.js	renderPhotoHistory	287
web/features.js	setActiveHistoryCard	312
web/features.js	openHistoryEntry	320
web/features.js	exitToMain	499
web/features.js	sprayAdvice	525
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web/features.js	setListenLabel	683
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web/field.js	setTab	309
web/field.js	renderPhiBanner	326
web/field.js	applyPlotLand	371
web/field.js	renderDiagnoseFieldPicker	385
web/field.js	wireDiagnoseFieldPicker	418
web/field.js	renderPlots	459
web/field.js	renderCalendar	543
web/field.js	renderOffline	583

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web/field.js	showOfflineDetail	609
web/field.js	renderAll	639
web/field.js	wireForm	646
web/field.js	wireTabs	664
web/field.js	init	674
web/geo.js	save	14
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web/geo.js	fromBrowser	31
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web/i18n.js	t	1504
web/i18n.js	govKey	1514
web/i18n.js	govField	1518
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web/mobile.js	setupHeaderNav	99
web/mobile.js	paneFromHash	104
web/mobile.js	wrapScrollShell	111
web/mobile.js	init	129
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web/notifications.js	isBn	12
web/notifications.js	esc	16
web/notifications.js	loadRead	21
web/notifications.js	saveRead	27
web/notifications.js	isRead	30
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web/notifications.js	mergeFeed	130
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web/offline-infer.js	isOtherCrop	55
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web/offline-infer.js	tensor	176
web/offline-infer.js	runModel	180
web/offline-infer.js	ensureLoaded	189
web/offline-infer.js	adviceFor	220

File	Function	Lines
web/offline-infer.js	otherMessage	227
web/offline-infer.js	predict	234
web/offline-infer.js	chatOffline	366
web/ops.js	esc	6
web/ops.js	toast	12
web/ops.js	formatDetail	21
web/ops.js	xhr	36
web/ops.js	money	54
web/ops.js	bindPasswordToggles	58
web/ops.js	authFail	74
web/ops.js	bootChrome	98
web/pwa.js	isStandalone	6
web/pwa.js	dismissKey	13
web/pwa.js	hideBanner	17
web/pwa.js	showBanner	21
web/shop.js	T	33
web/shop.js	isBn	37
web/shop.js	districtLabel	42
web/shop.js	upazilaLabel	48
web/shop.js	placeDisplay	54
web/shop.js	esc	72
web/shop.js	xhrGet	80
web/shop.js	xhrJson	92
web/shop.js	getSession	108
web/shop.js	authToken	113
web/shop.js	sessionUser	118
web/shop.js	isLoggedInIn	123
web/shop.js	isBn	127
web/shop.js	productName	132
web/shop.js	loadCart	137
web/shop.js	saveCart	144
web/shop.js	cartCount	148
web/shop.js	updateCartBadge	154
web/shop.js	playAddToCartAnim	169
web/shop.js	showToast	195
web/shop.js	setDrawerStep	206
web/shop.js	openCart	222
web/shop.js	closeCart	238
web/shop.js	setView	253
web/shop.js	applyShopWarnings	268
web/shop.js	setLocationStatus	301
web/shop.js	applyNearestUpazila	315
web/shop.js	fetchNearest	353
web/shop.js	detectLocation	376
web/shop.js	onErr	383
web/shop.js	typeLabel	412
web/shop.js	loadSuppliers	419
web/shop.js	rememberProduct	461
web/shop.js	loadProducts	465
web/shop.js	productInCart	528
web/shop.js	markAddButton	537
web/shop.js	addToCart	544
web/shop.js	setQty	562
web/shop.js	renderCart	574
web/shop.js	fillDistrictSelect	624
web/shop.js	fillCheckoutUpazila	647
web/shop.js	fillBrowseUpazila	666
web/shop.js	refillLocationSelects	686
web/shop.js	profileComplete	698
web/shop.js	setFormValues	702
web/shop.js	renderDeliverySummary	711
web/shop.js	refreshProfile	737
web/shop.js	goCheckout	780

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web/shop.js	readCheckoutFields	786
web/shop.js	checkout	809
web/shop.js	place	847
web/shop.js	loadOrders	888
web/shop.js	loadRecommendations	918
web/shop.js	renderRecommendInto	931
web/shop.js	setActiveCat	972
web/shop.js	addSku	984
web/shop.js	hydrateCartProducts	1003
web/shop.js	init	1020

Chapter 13

Counts

Python functions (agrosan + backend + scripts)	434
Python classes	25
JavaScript functions (<code>web/*.js</code>)	425
Largest Python files	<code>backend/main.py</code> 72, <code>shop_db.py</code> 44, <code>llm_chat.py</code> 39
Largest JS files	<code>shop.js</code> 58, <code>app-extra.js</code> 51, <code>field.js</code> 41

If a lecturer asks “how many functions does the project have?”: **about 860 in application code**, plus HTML-inline vendor/admin scripts. Do not count **site-packages**.